

Course Code: CS1885

Name of the Programme: BCA

Name of the Course: Set Theory

Semester: 2

Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
2	1+1+0	15L+15T

Course Objectives:

- To introduce foundational concepts in set theory including notation, operations, and representations.
- To develop analytical skills in understanding and applying relations and functions.
- To enhance the ability to model logical structures using combinatorics and pigeonhole principle.
- To provide essential mathematical tools for applications in computer science and data structures.

Course outline (Syllabus of the course)

Syllabus:

Module – 1 : Basic Concepts of Set Theory	No. of hours
	3
Introduction to sets, Set-builder notation, Types of sets (finite, infinite, singleton, null, equal, subset, universal), Cardinality.	

Module – 2: Set Operations and Venn Diagrams	No. of hours
	3
Operations on sets: Union, Intersection, Difference, Complement, Symmetric Difference Properties and laws of set operations, Venn diagrams, Cartesian product of sets.	

Module – 3: Relations	No. of hours
	3
Definition of relation, Types of relations: Reflexive, Symmetric, Transitive, Equivalence, Representation of relations, Matrix and digraph representation.	

Module – 4: Functions and Their Properties	No. of hours
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	3
Definition of functions, One-to-one and onto functions, Function composition, Invertibility and inverse of functions, Pigeonhole Principle.	

Module – 5: Permutations and Combinations	No. of hours
	3
Rule of Sum and Product, Permutations and Combinations, Binomial theorem, Combination with repetition.	

Course outcomes
a) Understand and apply basic concepts and operations of set theory using set notation and Venn diagrams.
b) Analyse and represent different types of relations using matrices and digraphs.
c) Demonstrate the ability to work with functions, their compositions, inverses, and apply the pigeonhole principle.
d) Apply basic counting techniques and solve problems involving permutations, combinations, and binomial expansion.

Text Books	
1	G. Shanker Rao, Discrete Mathematical Structures, New Age International, 2007 ; ISBN, 8122426697, 9788122426694
2	N Ch S N Iyengar, V M Chandrasekaran, Discrete Mathematics, Vikas Publishers, First Edition, 2003, ISBN :978-8125913627

Reference books	
1	Michael Huth & Mark Ryan, “Logic in Computer Science: Modelling and Reasoning about Systems”, Cambridge University Press, Second 2004, ISBN-(13):978-0521543101
2	Basavaraj S Anami and Venakanna S Madalli “Discrete Mathematics – A Concept based approach”, Universities Press, 2016, ISBN-(13): 978-8173719998
3	Lipschitz, Lipson, Discrete Mathematics, McGrawHill, Second 1992, ISBN : 978-0070380318
4	Rosen K, Discrete Mathematics and Its Applications, McGrawHill, 8th edition, 2021, ISBN: 978-9390727353

Tutorials	
1	Module 1: 3 Sessions
2	Module 2: 3 Sessions
3	Module 3: 3 Sessions
4	Module 4: 3 Sessions
5	Module 5: 3 Sessions